

FREE PILOT PROPOSAL

Atlas OT Scout

Understand what is really on your OT segment before you sign off.

An offline-first Android assessment instrument for water and wastewater operations: passive first, evidence reviewed by a human, bounded active identity only when explicitly approved.

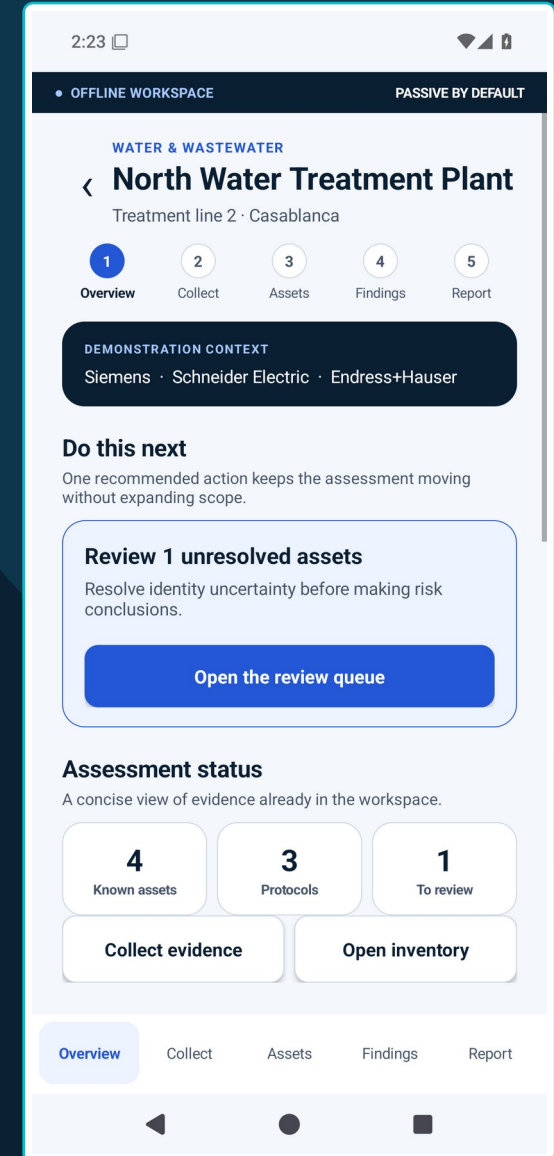
PASSIVE FIRST

OFFLINE FIRST

NO CONTROL WRITES

Pilot one bounded treatment or pumping segment at no cost - then decide based on evidence.

Executable prototype / API 35 evidence



Actual Android application

Why this project exists

Water OT teams need a current picture of a segment without turning the assessment itself into an operational risk.

1

Baseline drift

Spreadsheets and diagrams age. What is documented and what is actually communicating can diverge.

2

Intrusive discovery is hard to justify

Availability and process constraints make broad scanning a poor default in many control environments.

3

A device list is not a defensible handoff

The assessor still needs provenance, confidence, limitations and explicit review before conclusions are accepted.

The field question is not “how many devices did we scan?” It is “what can we prove about this exact segment?”

What Atlas OT Scout is

A field assessment workflow built around evidence quality, human review and explicit scope.

THE PROJECT

An offline-first Android field instrument for authorized OT asset assessment.

- One case = one site, one process area and one bounded control segment.
- Collect the least intrusive evidence that can close a real decision gap.
- Keep raw evidence, observations, accepted assets and findings as separate layers.

SCOPE

Site, process area, authorization and stop conditions

EVIDENCE

Passive capture/import first; exact active identity only when approved

REVIEW

Analyst acceptance before inventory or finding changes

HANDOFF

Reconciled inventory, evidence-linked findings and explicit remaining gaps

Not a penetration test. Not a certification audit. Not continuous monitoring.

What we have today

The prototype is already executable end-to-end; the pilot is for physical field qualification and customer value, not for validating a slide concept.

Current executable stack

CASE APP

Site / scope / review / inventory / findings

No Android INTERNET permission

CAPTURE BROKER Bounded passive PCAP stream

Receive-only software boundary

ISOLATED PARSER Protocol observations from sealed evidence

No network or case-store authority

NETWORK BROKER Signed one-use active identity operation

Compiled Modbus identity only

Raw evidence → proposed observations → reviewed assets → findings → readiness

5

guided assessment stages

4

sourced passive protocol fixtures

1

bounded Modbus identity request



Android API 29 + API 35 guided UI journeys

site → collect → assets → findings → report



PCAP / PCAPNG analysis with hashes and provenance

Modbus/TCP, DNP3, IEC-104 and BACnet sourced fixtures



Receive-only native capture path exercised in CI

bounded AF_PACKET capture and zero-transmission assertion

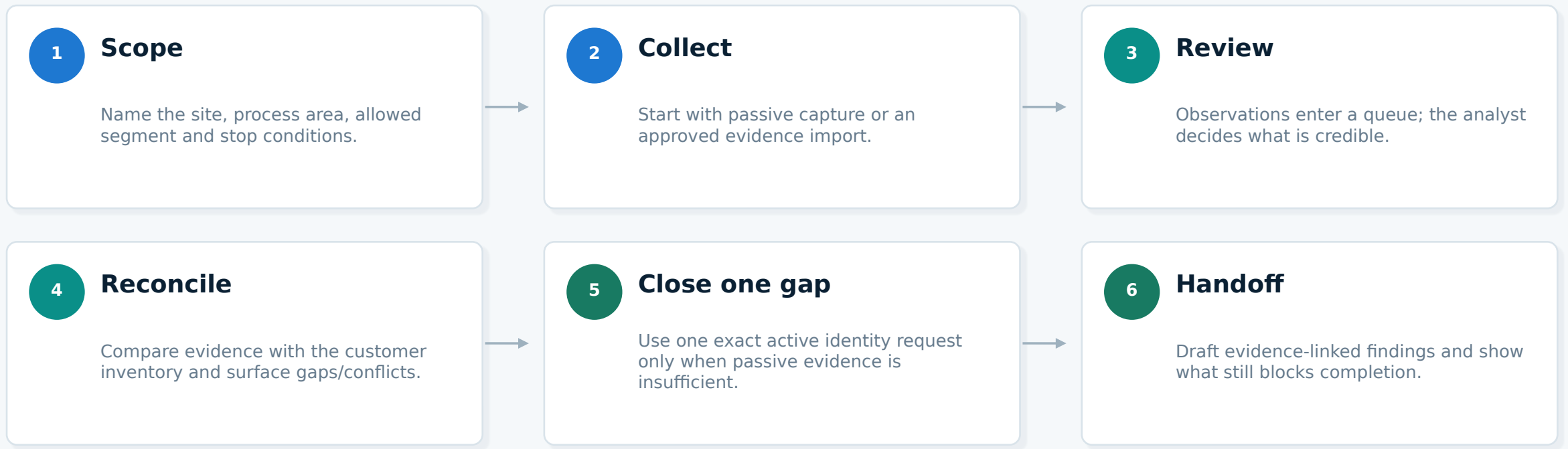


Signed UI → broker → TCP → evidence journeys

PyModbus identity plus conservative Modbus-TK / Conpot outcomes

How an assessment works

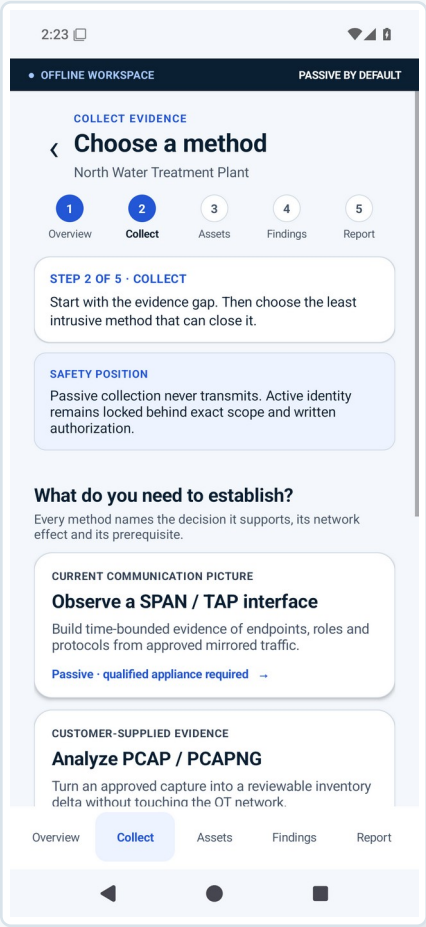
The assessor follows one controlled journey from scope to a decision-ready handoff.



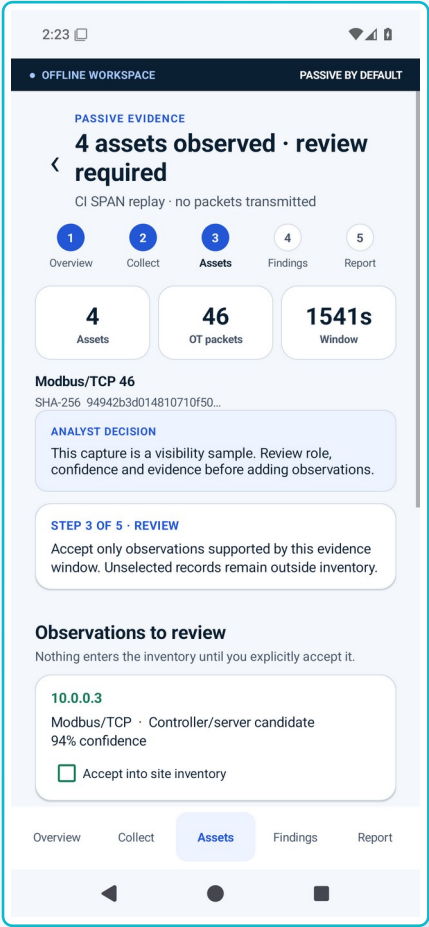
The product does not treat “observed” as “true.” Every important state change keeps the evidence and the analyst decision visible.

It starts passive

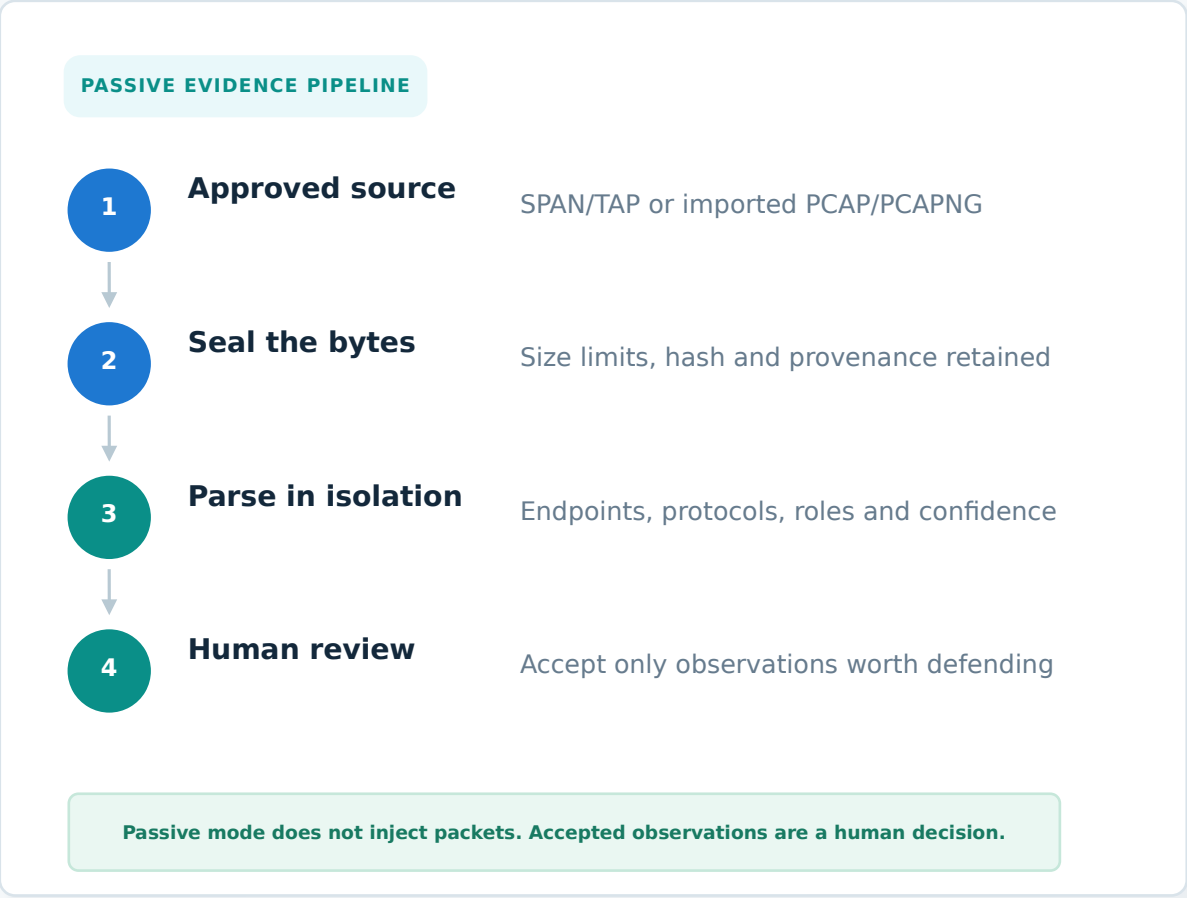
Use an existing capture or an approved SPAN/TAP window before asking the network to answer anything.



Choose the least intrusive method



Review observations before accepting them



When passive is not enough, active behavior stays narrow

The current active function is one exact Modbus device-identification request to one approved target - and it fails closed on scope.

2:23

AUTHORIZED ACTIVE CHECK

< **Identify one Modbus device**

North Water Treatment Plant

1 **Work order**

Case reference

PO-WATER-001

Process area

Treatment line 2

2 **Exact target and scope**

Target controller IPv4

192.0.2.5

Authorized IPv4 scope (CIDR)

10.0.2.0/24

Modbus unit ID

1

ACTIVE LIMITS

1 identity request · TCP/502 · 1.5 s timeout · no register reads or writes

3 **Authorization**

☒ I confirm written operational and security authorization for this target, scope and time window.

Target is outside the authorized CIDR. The app will not expand

Out-of-scope target is blocked locally

8:29

ACTIVE EVIDENCE

< **Controller identified**

E2E-WATER-DEMO · constrained check complete

IDENTITY CONFIRMED

MODBUS/TCP · 10.0.2.2:502

Interface: connected network

PyModbus · ATLAS-CI · 3.11.3

FC 43 / MEI 14 basic device identity response

Evidence bytes: 42

NEXT DECISION

Review the evidence before adding this asset. No register read or write was performed.

Add to asset inventory

Authorized target returns identity evidence

WHAT IT MAY DO

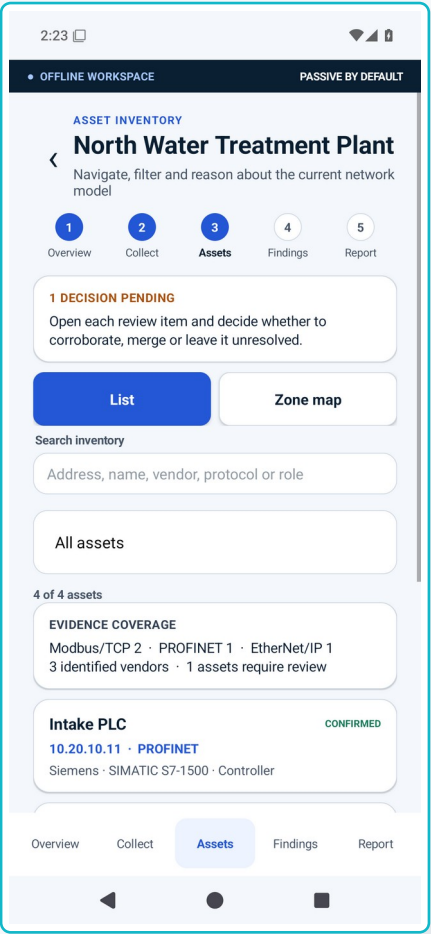
- One explicit IP target
- One explicit authorized CIDR
- One Modbus FC 43 / MEI 14 basic identity request
- Short timeout and bounded response
- Review the result before inventory changes

WHAT IT DOES NOT DO

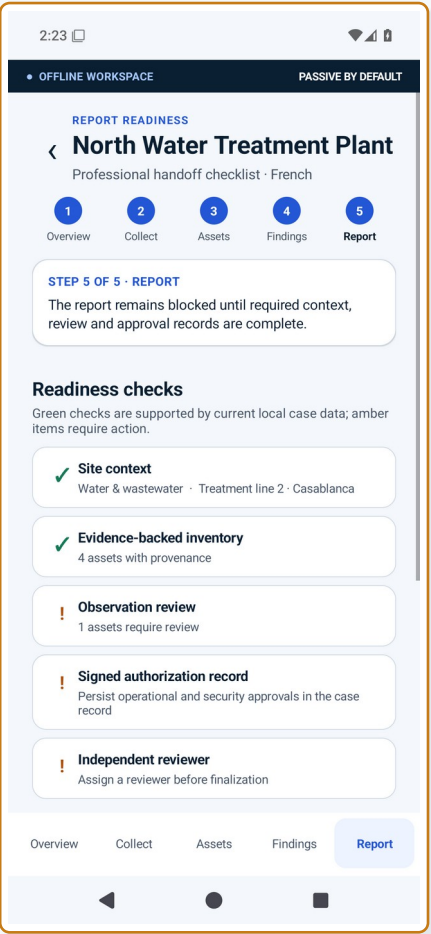
- No subnet or port scan
- No unit-ID sweep
- No register read or write
- No credentials or exploit checks
- No scope broadening to make a test pass

What the prospect gets from the pilot

Not a scanner export - a reviewable evidence pack that shows what changed, what is supported and what still needs work.



Reconciled asset view



Explicit readiness blockers

INVENTORY RECONCILIATION

Confirmed, probable, missing, unexpected and conflicting asset records.

EVIDENCE PROVENANCE

Capture/source hashes, collection context, endpoint observations and confidence.

DRAFT FINDINGS

Condition, affected asset, evidence references, confidence, limitations and recommended next action.

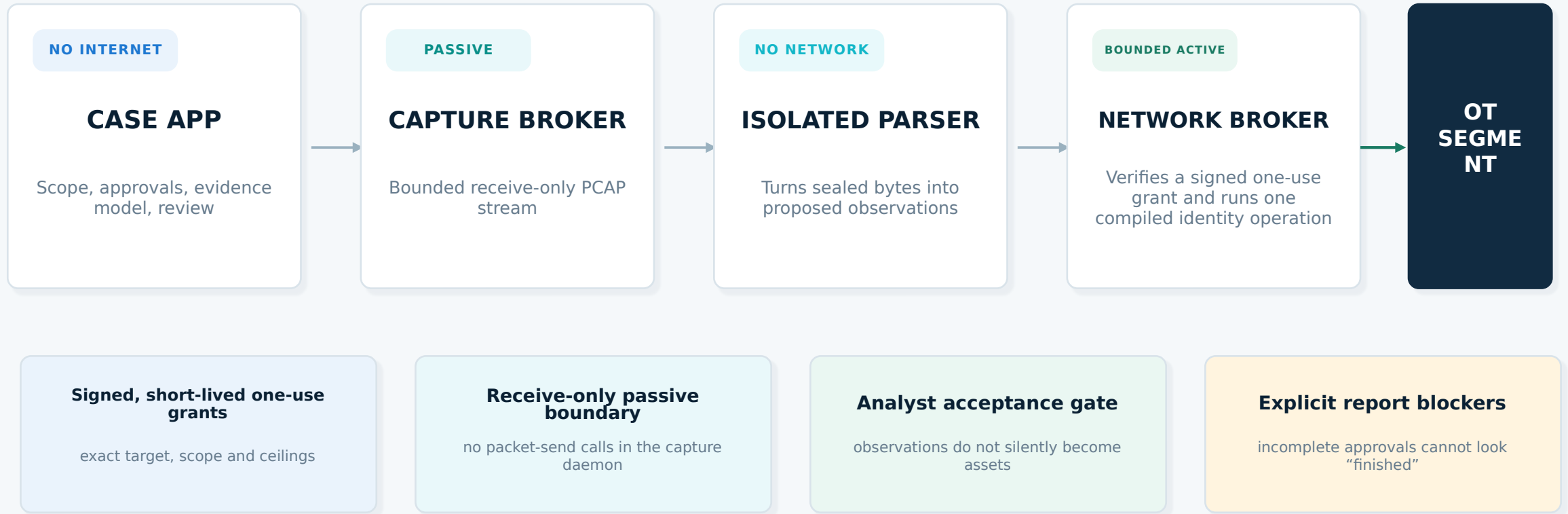
READINESS / GAPS

What is defensible now, what remains unresolved and what is needed before a formal handoff.

The current prototype produces these views and evidence states. Production signing/export packaging is still a release gate.

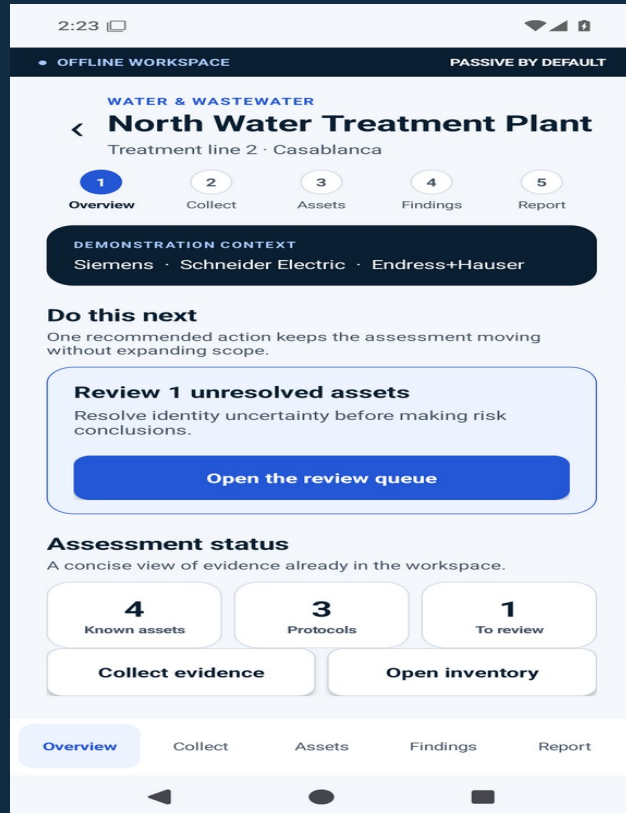
Designed to fail closed

Network and parsing authority are separated so the general application cannot silently become an unrestricted network tool.



See the real workflow

This is the running Android application, recorded continuously - not a screenshot animation. In PowerPoint, click the video to play it.



▶ **PLAY 39-SECOND DEMO**

What to watch for

- Passive collection before active behavior
- Analyst accepts observations before inventory changes
- Out-of-scope target is rejected
- One authorized Modbus identity request
- Report remains blocked while required records are missing

WHY THIS MATTERS

The demo shows behavior that directly supports the pilot conversation:

- The tool begins from an approved assessment context.
- It is not “scan first, explain later.”
- The safety boundary is visible to the operator.
- The handoff keeps limitations visible instead of hiding them.

What is proven today - and what the field pilot must prove

The prototype has software evidence. The missing evidence is physical and operational, which is exactly why a bounded pilot is the next rational step.

PROVEN IN CI / EMULATION

- Guided site → collect → assets → findings → report workflow
- PCAP/PCAPNG hashing, parsing and review-first inventory behavior
- Out-of-scope active request rejected before execution
- Signed one-target Modbus identity path to independent testbeds
- Receive-only software capture boundary exercised and zero-TX asserted
- Conservative service-only outcomes when full identity is unsupported

FIELD PILOT MUST PROVE

- Physical Android / USB-Ethernet / SPAN-TAP compatibility
- Packet loss, timing and stability on real plant traffic
- Actual protocol and device mix at the selected water segment
- Operator and assessor workflow fit under site constraints
- Whether reconciliation identifies valuable gaps or validates the baseline
- Which report/export format the customer considers useful and defensible

Proposed free pilot: one bounded water segment

No-fee field evaluation of the workflow and evidence value. No broad scan. No control writes. No obligation to continue.

BEFORE

Scope & authorize

Choose one treatment or pumping segment, agree the evidence window, stop conditions and approved methods.

ON SITE

Collect safely

Use passive evidence first. Run exact one-target identity checks only where the customer explicitly approves them.

AFTER

Review together

Reconcile the customer inventory, review evidence-linked gaps/findings and agree what would be needed for a production deployment.

What we need from you

- One approved control segment + technical contact
- SPAN/TAP access or an approved capture set + current inventory extract

What you get

- Pilot evidence pack + reconciled inventory + prioritized gaps
- Joint debrief: what is proven, what remains uncertain and whether a rollout is worth pursuing

THE DECISION

Do one bounded pilot. Then decide from the evidence.

A successful pilot should answer five questions:

SAFE

Did the assessment stay inside the approved methods and scope?

VISIBLE

Did the evidence give a useful communication picture of the segment?

VALUABLE

Did reconciliation expose a real gap, conflict or useful baseline confirmation?

DEFENSIBLE

Can each important claim be traced back to evidence and analyst review?

USABLE

Would operations/security teams use this workflow again?

Next step

- 1 Select one segment
- 2 Nominate the technical contact
- 3 Agree the evidence window
- 4 Run the free pilot
- 5 **Decide whether to continue**

PILOT FIRST. BUY LATER.